Robust Joint Motion Forecasting Under Missing and Noisy Observations

Compact MotionLM-style joint forecaster on the Waymo Open Motion Dataset — corruption-aware training and bounded RL post-training

Status: FINAL (locked held-out set, 3 seeds). Development numbers use manifests/dev.json; held-out numbers use the locked manifests/final.json.

Research question and motivation

Question

Does a joint two-agent autoregressive forecaster keep its accuracy and physical consistency when the observed history is incomplete or noisy — and can training on corrupted observations, plus bounded reward post-training, make it more robust without hurting clean accuracy?

Why it matters

Deployed trackers drop frames and jitter. Forecasters trained only on clean logs can degrade sharply; robustness must be measured with the same evaluator on the same scenes, with uncertainty, before claiming an improvement.

Approach

Reproduce a compact MotionLM-style joint model (M0), train a matched corruption-aware variant (M1) and an independent-agent ablation, then post-train M1 with REINFORCE + KL + replay against simpler controls. Preregistered thresholds; paired group bootstrap.

- Preregistered success rule (PLAN.md): ≥ 5 % relative reduction of core-suite joint minADE@6, ≤ 3 % clean worsening, ≤ 0.5 pp overlap increase
- Scope kept to forecasting, robustness and bounded RL — no observation selection, no architecture expansion

Task and dataset

9,144
training windows (9,144 after removing scenes that share a source segment with dev)

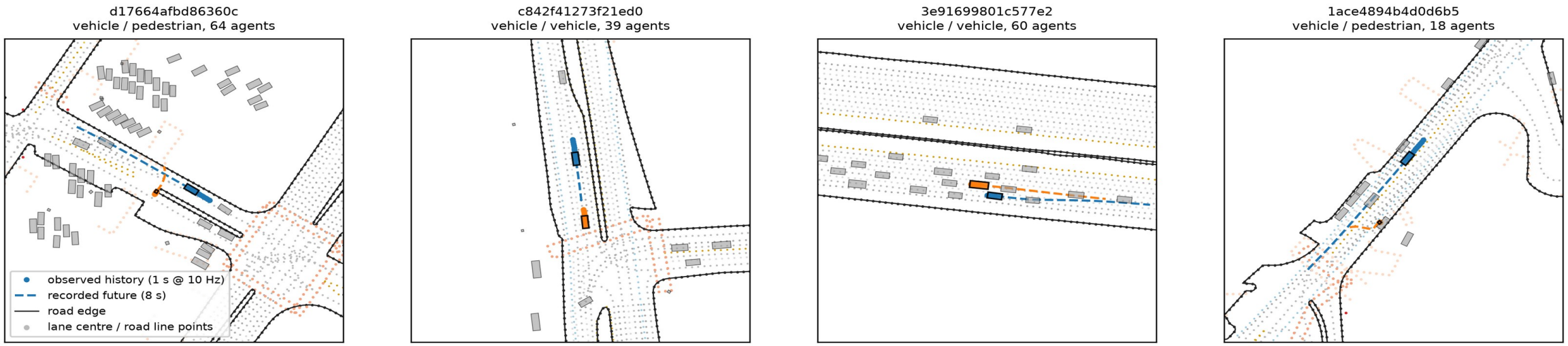
500
development scenes (selection, tuning, sampling freeze)

1,000
final held-out scenes, LOCKED (official validation shards)

1 s → 8 s
history @ 10 Hz → 16 joint steps @ 2 Hz

2
interacting targets per scene (objects_of_interest)

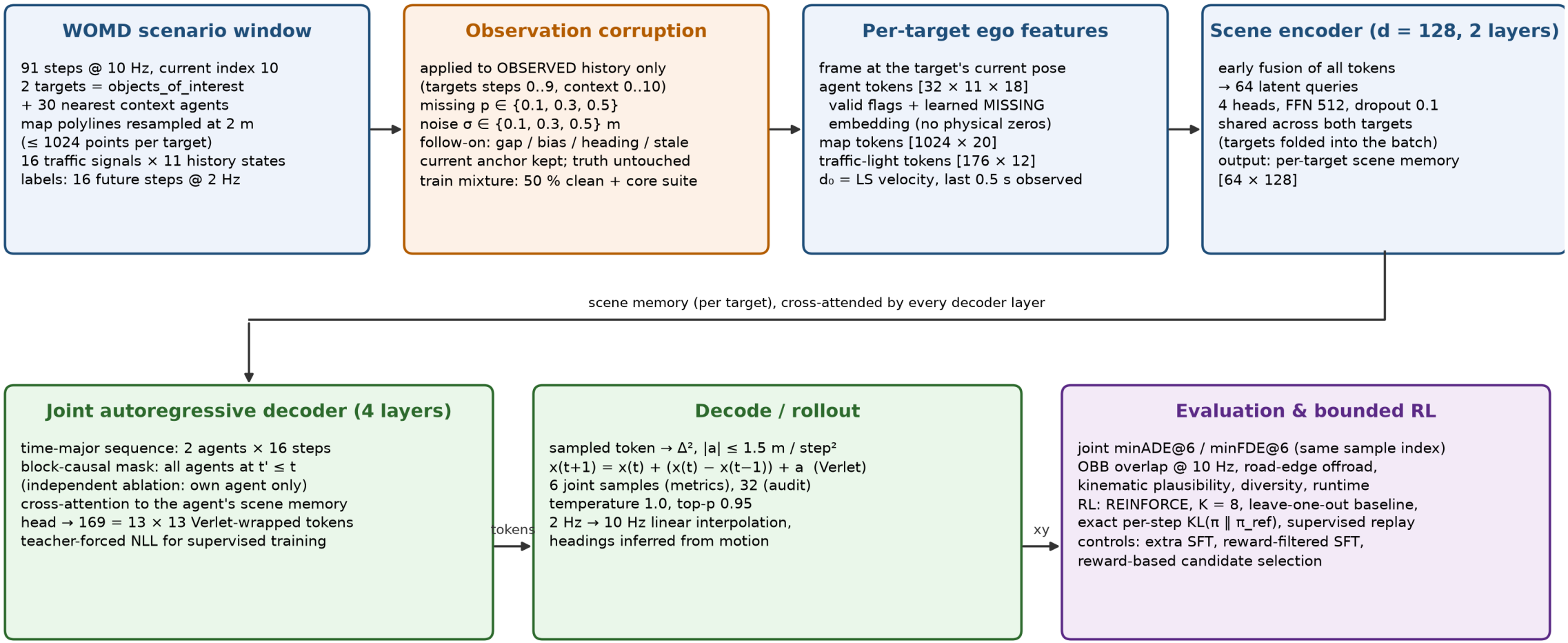
Waymo Open Motion Dataset v1.3.0 — interactive pairs (blue, orange) with map and context agents (grey)



- WOMD v1.3.0 scenario format, 48 training + 12 validation shards; 1.7 % of pair scenarios excluded by the 10 Hz timing rule (dropped frames, not interpolated)
- Calibration on dev: tokenizer saturation 0.08 %, reconstruction 0.092 m; offroad convention validated by log playback (0.98 % offroad); logged-future overlap reference 1 % on jointly valid intervals (audit v2)

Implemented architecture

Compact MotionLM-style joint forecaster — 1.79 M parameters (as implemented in src/waymo_av/model.py)



Experimental design and baselines

Method	Training observations	Role
Constant position / constant velocity	— (deterministic, 1 sample)	non-learned floor; CV uses the model's own d_0 estimator
Independent decoder (Mind)	clean	ablation: same model, no cross-agent attention
Joint M0	clean	clean-trained baseline (reference for all paired deltas)
Joint M1	50 % clean + core corruption suite	corruption-aware model
M1 + extra SFT	as M1, matched extra steps	control: more supervised training
M1 + reward-filtered SFT	own samples, reward-filtered	control: reward signal without policy gradient
M1 + bounded RL	REINFORCE, K = 8, LOO baseline, KL, replay	post-training (reward: overlap, offroad, kinematics)

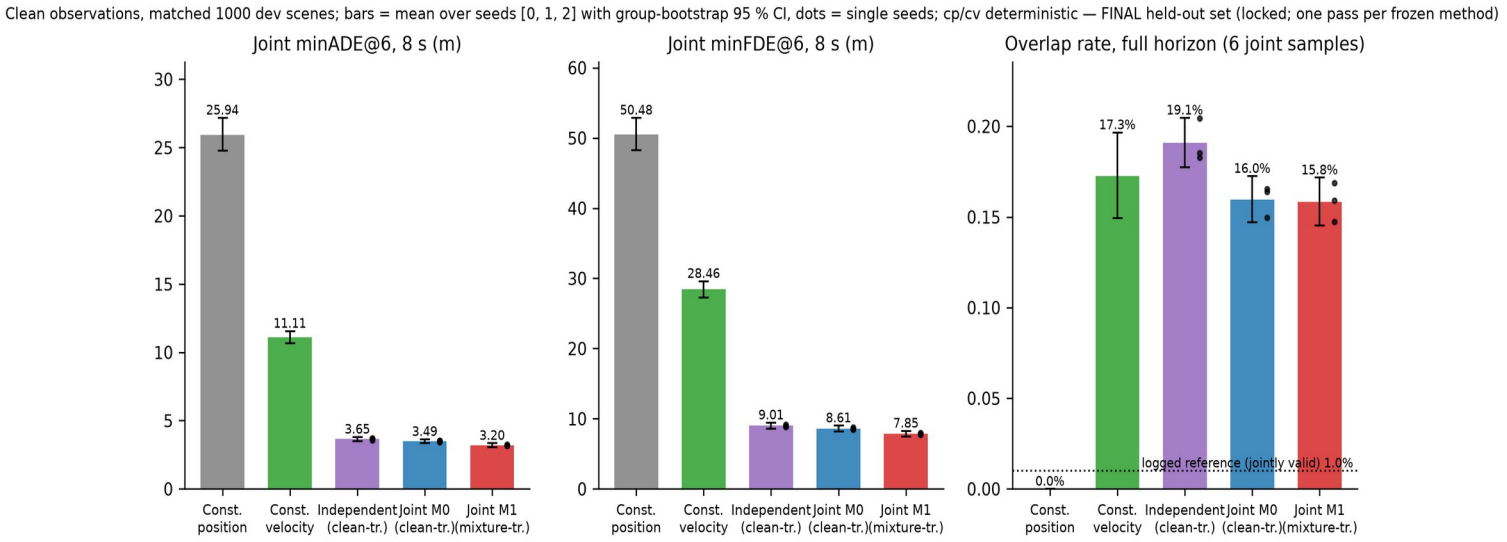
Corruption suite (observations only)

Core: missing 10/30/50 % of history frames; Gaussian position noise σ = 0.1/0.3/0.5 m.
Held-out severities: missing 70 %, σ = 1.0 m, compound.
The current anchor frame is kept; ground truth is never corrupted.

Measurement

Joint minADE@6 / minFDE@6 over the same sample index (6 samples; 32-sample audit for overlap/diversity).
OBB overlap at 10 Hz, road-edge offroad (vehicles), kinematic plausibility, diversity.
Paired bootstrap over heuristic source-segment groups; 3 seeds; checkpoints chosen on dev only; final set evaluated once after freeze.

Clean forecasting results

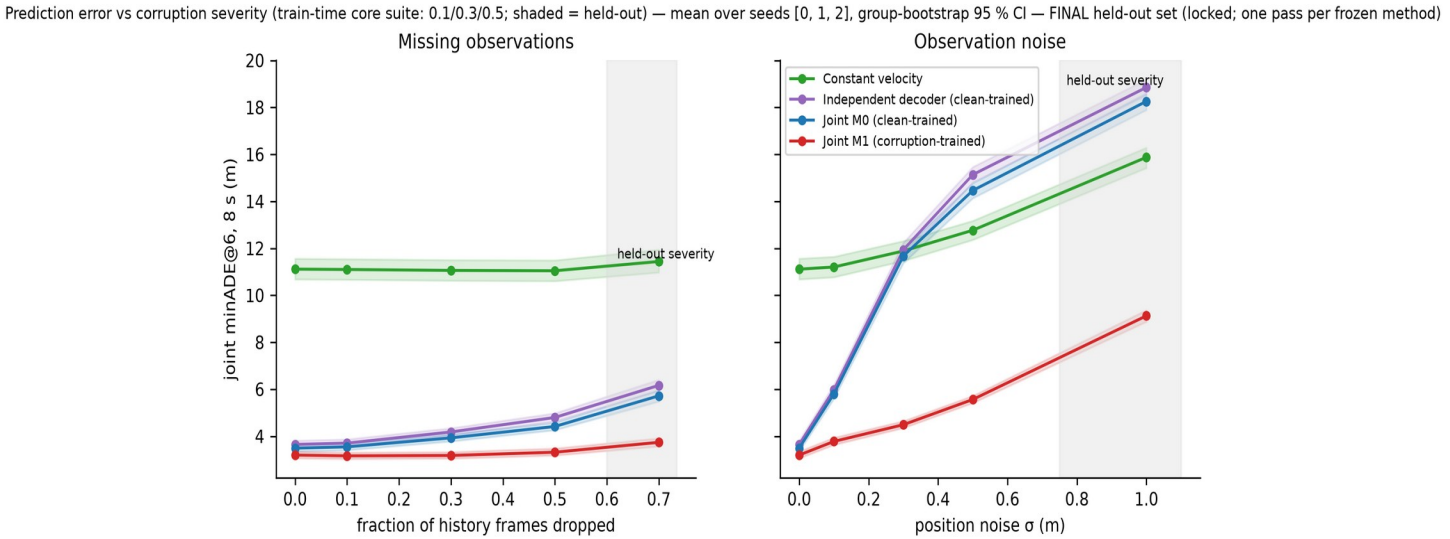


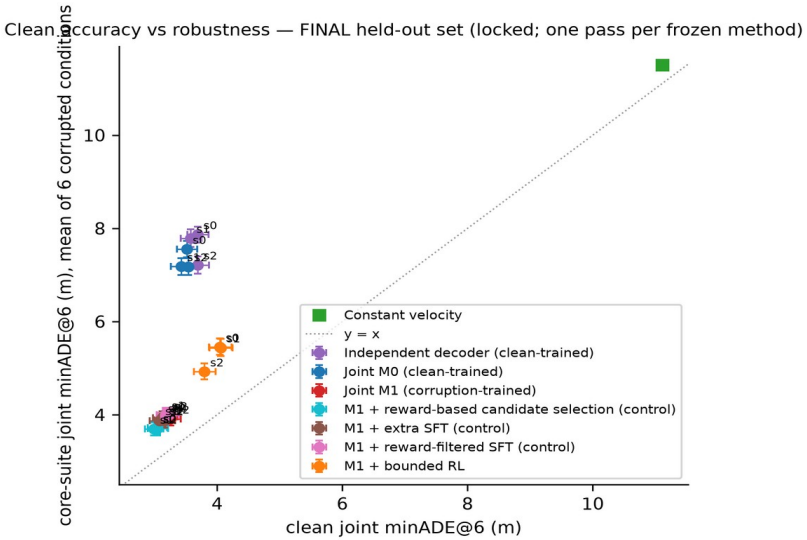
Joint minADE@6 (m), clean, seeds 0,1,2

Constant velocity: 11.11 [10.68, 11.56]
Independent: 3.65 [3.51, 3.81]
Joint M0: 3.49 [3.35, 3.64]
Joint M1: 3.20 [3.06, 3.35]
Joint − independent: -0.16 [-0.24, -0.08] (-4.3 %)

- Evaluated on the FINAL held-out set (1,000 locked scenes); identical scenes and sampling (6 joint samples, T = 1.0, top-p 0.95) for every learned model; cp/cv are deterministic
- Overlap, clean, 32-sample audit: M0 15.9 [14.7, 17.1] % over the full generated horizon; on jointly valid intervals M0 15.7 [14.6, 16.9] % vs logged reference 1.0 [0.4, 1.6] % (same approximate test applied to logs; not a lower bound)
- Checkpoints selected on dev by their own criterion (M0: clean jminADE; M1: mean of clean and suite); 12k steps, 1.79 M parameters, d = 128

Robustness to missing and noisy observations



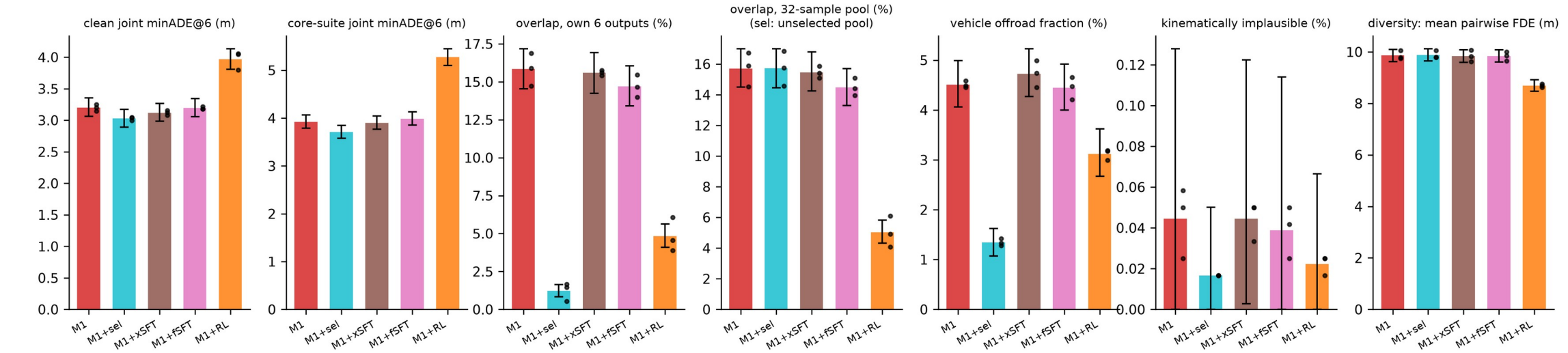


seed	core-suite rel. Δ (≤ −5 %)	clean rel. Δ (≤ +3 %)	core overlap Δ pp (≤ +0.5)	verdict
0	-47.7 % [-49.5, -45.9]	-9.0 % [-13.0, -5.3]	-0.26 [-1.16, +0.62]	PASS
1	-45.8 % [-47.6, -43.8]	-8.2 % [-11.8, -4.6]	+1.72 [+0.77, +2.67]	FAIL
2	-45.5 % [-47.7, -43.4]	-8.2 % [-11.8, -4.6]	+0.06 [-0.91, +1.08]	PASS

M1 vs M0 (paired, group bootstrap)

Core suite jminADE: -3.39 [-3.50, -3.28] (-46.4 %)
Held-out suite: -5.46 [-5.63, -5.29] (-47.9 %)
Clean: -0.30 [-0.38, -0.22] (-8.5 %)
Core overlap: +0.51 [-0.19, +1.19] pp

Bounded RL post-training and controls



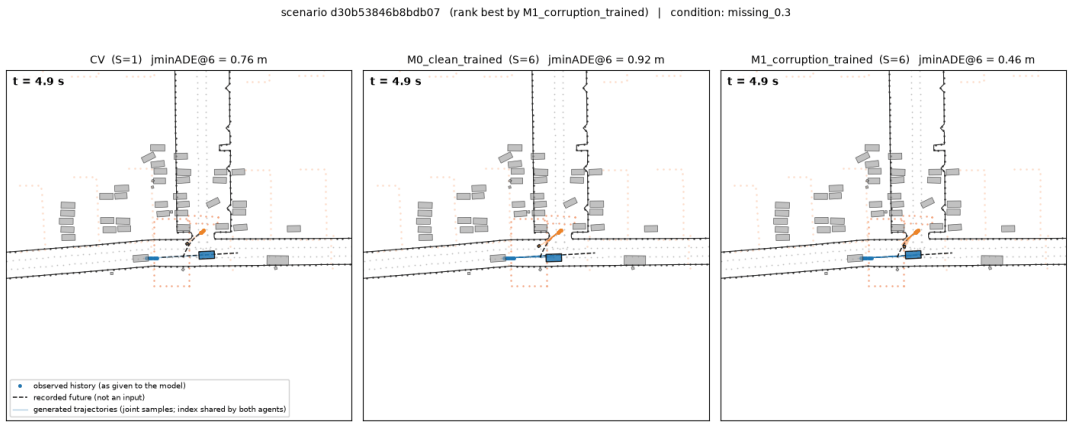
RL vs its M1 initialisation (clean, paired)

Clean jminADE@6: +0.77 [+0.67, +0.86] (+24.0 %)
Overlap, 32 samples: -10.67 [-11.60, -9.80] pp
Offroad: -1.39 [-1.63, -1.16] pp
Diversity, pairwise FDE: -1.17 [-1.29, -1.05] m
Excessive stopping: RL 3.49 % vs M1 0.43 % (dev diagnostic)
  ends < 1 m when logged motion > 5 m, per agent-sample
Frozen rule: does not count in seeds 0, 1, 2
  needs: beat both SFT controls on overlap, clean Δ ≤ +3 %

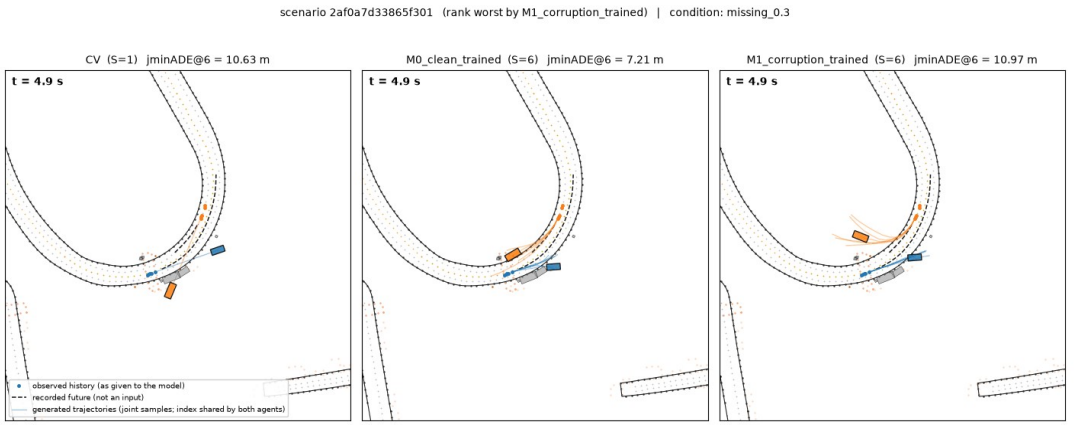
Controls vs M1

Reward-filtered SFT, overlap32: -1.23 [-1.53, -0.91] pp
Extra SFT (matched steps), overlap32: -0.26 [-0.56, +0.03] pp
Reward-based selection, best 6 of 32:
  own-output overlap -14.62 [-15.80, -13.52] pp
  clean jminADE -0.17 [-0.22, -0.12] (-5.4 %)
  cost: 32 rollouts + reward scoring per scene
Reward = approximate geometric consistency, not verified safety

Qualitative examples and failures



Success case — scenario d30b53846b8bdb07, condition missing_0.3 — CV 0.76 m, M0_clean_trained 0.92 m, M1_corruption_trained 0.46 m



Failure case — scenario 2af0a7d33865f301, condition missing_0.3 — CV 10.63 m, M0_clean_trained 7.21 m, M1_corruption_trained 10.97 m

- Selection rule (not hand-picked): pool = first 64 manifest scenes; select 'mixed' by M1_corruption_trained joint minADE@6; identical scenes/corruption/sampling across methods
- Each panel shows the identical scene, corruption realisation and sampling; dots = observed history as given to the model, black dashed = recorded future (never an input), coloured = generated joint samples with one shared sample index for both agents
- Animations (GIF + MP4, synchronized, fixed camera): reports/demos/<set>/<scenario>_<condition>.{gif,mp4}

FINAL (locked held-out set, 3 seeds)

Conclusions, limitations, remaining work

- Clean: joint M0 3.49 [3.35, 3.64] m vs CV 11.11 [10.68, 11.56] m; joint − independent -0.16 [-0.24, -0.08] (-4.3 %)
- Robustness: M1 − M0 on the core suite -3.39 [-3.50, -3.28] (-46.4 %); clean -0.30 [-0.38, -0.22] (-8.5 %); verdicts: seed 0 PASS, seed 1 FAIL, seed 2 PASS
- RL (mixed/negative): overlap -10.67 [-11.60, -9.80] pp vs M1 but clean jminADE +0.77 [+0.67, +0.86] (+24.0 %), diversity -1.17 [-1.29, -1.05] m, excessive stopping 3.49 % vs 0.43 %; frozen rule: seed 0 DOES NOT COUNT, seed 1 DOES NOT COUNT, seed 2 DOES NOT COUNT; the best-6-of-32 selection control dominates RL on every axis at 32-sample cost
- Limitations: compact model and ~9k training scenes (not a leaderboard reproduction); official metrics package unavailable (custom metrics audited on oracles); the 2 Hz box-overlap test is approximate (logged reference 1 % on jointly valid intervals; RL reward = approximate geometric consistency, not verified safety); source-segment grouping is heuristic; synthetic corruptions are stress tests, not sensor models
- Final held-out evaluation done once per frozen method; see report for full tables

Follow-up: initial-motion error and a cheap estimator fix

Protocol registered before comparisons · dev 500 scenes (exploratory) · one frozen confirmation pass on 141 unused validation scenes × 3 seeds

Mechanism test (oracle d0, identical tokens)

4–18 %

of M0's noise-induced excess error removed by oracle d0

- d0 = start velocity for integrating tokens; not a network input
- Most of M0's noise penalty enters via corrupted input features
- M1's tokens partly cancel d0 error (c_4 0.6–0.9 at 0.1 m)

Estimator benefit: anchored1s (confirmation)

−5.8 % M0 · −9.9 % M1

follow-up corrupted mean; confirmed in all 3 seeds

- Clean cost +1.3 % M0, +1.5 % M1 (pooled CIs exclude 0)
- Overlap +1.5 / +1.7 pp (adverse); offroad −2.3 / −2.1 pp
- No gain under 0.5 m bias; M1 slightly worse at 0.1 m noise

Remaining M1 advantage (same estimator)

−5.76 m (−53 %)

M1 − M0 with anchored1s, follow-up corrupted mean; confirmed

- With the v1 estimator: −5.87 m (−51 %)
- Gap change +0.11 m [−0.02, +0.25]: directionally consistent, not confirmed
- Not evidence of mediation; no model was retrained

Confirmation limits

- 141 scenes, not the planned 500 (shard download needed reauth)
- The same 141 scenes are used for all 3 seeds
- Source separation is heuristic; 37 stationary-SDC scenes untestable
- Frozen v1 checkpoints; synthetic corruption with an exact current frame (favours the anchored fit)
- Follow-up suite (noise 0.1–1.0 m, bias 0.5 m) is not the v1 core suite